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Department of Mathematics, Bioinformatics and Computer Applications
Mathematics-IV (MTH-241)

Assignment-2

Due Date:

Even Sem: 2024-25

Q.1. The ABC car company has warehouses in Calcutta (W_1), Patna (W_2), and Bhavnagar (W_3) and markets in New Delhi (M_1), Dhanbad (M_2) and Calicut (M_3). At a particular time, the company has 61 cars in Calcutta, 49 in Patna and 90 in Bhavnagar. The company plans to transport 52 cars to New Delhi, 68 to Dhanbad and 80 to Calicut. The transportation costs per unit (in rupees) as well as the above data are given in the following table as

		M_1	M_2	M_3	Availability
Origins	W_1	26	23	10	61
	W_2	14	13	21	49
	W_3	16	17	29	90
Requirements		52	68	80	

Find the initial feasible solution by row minima (RM), column minima (CM), Least cost method (LCM), North West Corner rule (NWCR) and Vogel's approximation method (VAM). What do you infer about these methods? Write these methods in hierarchical order of accuracy. Also, determine the values of RM-CM, LCM-CM, NWCR-LCM and VAM-LCM.

Q.2. A company has 4 warehouses and 6 stores. The cost of shipping one unit from warehouse i to j is c_{ij} . If

$$C = \begin{bmatrix} 7 & 10 & 7 & 4 & 7 & 8 \\ 5 & 1 & 5 & 5 & 3 & 3 \\ 4 & 3 & 7 & 9 & 1 & 9 \\ 4 & 6 & 9 & 0 & 0 & 8 \end{bmatrix}$$

and the requirements of the six stores are 4, 4, 6, 2, 4, 2 and quantities available at the warehouses are 5, 6, 2, 9. Find the minimum TP cost of shipping by LCM.

Q.3. Obtain the initial basic feasible solution and optimal solution for the following transportation problem using (a) Column minima method (b) Row minima method.

		A	B	C	Availability
Origins	U	2	7	4	5
	V	3	3	1	8
	w	5	4	7	7
	X	1	6	2	14
Requirements		7	9	18	

Which optimal solution is better than Vogel's approximation method? Describe in details.

Q.4. A company has three plants at locations A, B, and C, which supply to warehouses located at D, E, F, G and H. Monthly plant capacities are 800, 500, and 900 units respectively. Monthly warehouse requirements are 400, 400, 500, 400, and 800 units respectively. Unit transportation costs (in Rs) are given as

	D	E	F	G	H
A	5	8	6	6	3
B	4	7	7	6	5
C	8	4	6	6	4

Determine an optimum distribution by Vogel's approximation method for the company in order to minimize the total transportation cost.

Q.5. A leading firm has three auditors. Each auditor can work upto 160 hours during the next months, and during this time three projects must be completed. Project 1 will take 130 hours, project 2 will take 140 hours, the project 3 will take 160 hours. The amount per hour that can be billed for assigning each auditor to each project is given in the table

		1	2	3
Auditor	1	1200	1500	1900
	2	1400	1300	1200
	3	1600	1400	1500

Formulate this as a transportation problem and find the optimal solution. Also find the maximum total billings during the next month.

Q.6. Determine the initial feasible solution and optimal solution of the following transportation problem using the (i) North West Corner rule (ii) Least Cost method

		Destination				
		D ₁	D ₂	D ₃	D ₄	Availability
Origins	O ₁	6	4	1	5	14
	O ₂	8	9	2	7	16
	O ₃	4	3	6	2	10
Requirements		6	10	15	4	

Q.7. Find the basic feasible solution of the following transposition problem by the North West Corner rule.

	1	2	3	4	5	Availability
A	4	3	1	2	6	80
B	5	2	3	4	5	60
C	3	5	6	3	2	40
D	2	4	4	5	3	20
Requirements	60	60	30	40	10	

Q.8. A company manufacturing air coolers and has two plants located at Mumbai and Kolkata with a capacity of 200 units and 100 units per week, respectively. The company supplies the air coolers to its four show-rooms situated at Ranchi, Delhi, Lucknow, and Kanpur with have a maximum demand of 75, 100, 100 and 30 units respectively. Due to the differences in raw material cost and transportation cost, the profit per unit in rupees differs which is shown in table below

	Ranchi	Delhi	Lucknow	Kanpur
Mumbai	90	90	100	110
Kolkata	50	70	130	85

Plan the production programme so as to maximize the profit. The company may have its production capacity at both plants partly or wholly unused.

Q.9. Find the optimal solution to the following transportation problem using the Vogel's approximation method. The cells in the following table contain the transportation cost in rupees

		W₁	W₂	W₃	W₄	W₅	Availability
Requirements	F₁	7	6	4	5	9	40
	F₂	8	5	6	7	8	30
	F₃	6	8	9	6	5	20
	F₄	5	7	7	8	6	10
		30	30	15	20	5	

Q.10. Consider the following unbalanced transportation problem

		To			
		1	2	3	Supply
From	1	5	1	7	10
	2	6	4	6	80
	3	3	2	5	15
	Demand	75	20	50	

Since there is not enough supply, therefore some of the demands at these destinations may not be satisfied. Suppose, there are penalty costs for every unsatisfied demand units which are given by 5, 3 and 2 for destination 1, 2 and 3, respectively. Find the optimal solution.

Q.11. A product is produced by four factories A, B, C and D. The unit production costs for them are Rs.2, Rs.3, Rs.1 and Rs.5 respectively. Their production capacities are for factory A-50 units, B-70 units, C-30 units and D-50 units. These factories supply the product to four stores. The demands of these stores are 25, 35, 105 and 20 units, respectively. Units cost in rupees from each factory to each store is given in the table below

	1	2	3	4
A	6	4	1	5
B	8	9	2	7
C	4	3	6	2
D	6	10	15	4

Determine the extent of deliveries from each of the factories to each of the stores so that the total production and transposition cost is minimum.

Q.12. A company produces a small component for all industrial products and distributes it to five the wholesalers at a fixed delivered price of Rs. 2.50 per unit. Sales forecasts indicate that monthly deliveries will be 3000, 3000, 10000, 5000, 4000 units to wholesalers 1, 2, 3, 4 and 5 respectively. The monthly production capacities are 5000, 10000, and 12500 at plants 1, 2 and 3 respectively. The direct costs of production of each unit are Rs. 1, Rs. 0.90, and Rs. 0.80 at plants 1, 2 and 3, respectively. The transportation costs of shipping a unit from a plant to a wholesaler are given below.

		Wholesaler				
		1	2	3	4	5
Plant	1	0.05	0.07	0.1	0.15	0.15
	2	0.08	0.06	0.09	0.12	0.14
	3	0.1	0.09	0.08	0.1	0.15

Find how many components of each plant supplies to each wholesaler in order to maximize the profit.